

SECTION ONE

Unleash Your Creativity

CHAPTER 1

Reclaim Your Creative Confidence

by Tom Kelley and David Kelley

Most people are born creative. As children, we revel in imaginary play, ask outlandish questions, draw blobs and call them dinosaurs. But over time, because of socialization and formal education, a lot of us start to stifle those impulses. We learn to be warier of judgment, more cautious, more analytical. The world seems to be divided into *creatives* and *noncreatives*, and too many people consciously or unconsciously resign themselves to the latter category.

And yet we know that creativity is essential to success in any discipline or industry. According to an IBM survey of chief executives around the world, it's the most sought-after trait in leaders. No one can deny that creative thinking has enabled the rise and continued success of countless companies, from startups to stalwarts.

Students often come to Stanford University's "d.school" (which was founded by one of us—David—and is formally known as the Hasso Plattner Institute of Design) to develop their creativity. Clients work with IDEO, our design and innovation consultancy, for the same reason. But along the way, we've learned that our job isn't to *teach* them creativity. It's to help them *rediscover* their creative confidence—the natural ability to come up with new ideas and the courage to try them out. We do this by giving them strategies to get past four fears that hold most of us back:

- Fear of the messy unknown
- Fear of being judged
- Fear of the first step
- Fear of losing control

Easier said than done, you might argue. But we know it's possible for people to overcome even their most deep-seated fears. Consider the work of Albert Bandura, a world-renowned psychologist and Stanford professor. In one series of early experiments, he helped people conquer lifelong snake phobias by guiding them through a series of increasingly demanding interactions. They would start by watching a snake through a two-way mirror. Once comfortable with that, they'd progress to observing it through an open door, then to watching someone else touch the snake, then to touching it themselves through a heavy leather glove, and, finally, in a few hours, to touching it with their own bare hands. Bandura calls this process of experiencing one small success after another "guided mastery." The people who went through it weren't just cured of a debilitating fear they had assumed was untreatable. They also had less anxiety and more success in other parts of their lives, taking up new and potentially frightening activities like horseback riding and public speaking. They tried harder, persevered longer, and had more resilience in the face of failure. They had gained a new confidence in their ability to attain what they set out to do.

We've used much the same approach over the past 30 years to help people transcend the fears that block their creativity. You break challenges down into small steps and then build confidence by succeeding on one step after another. Creativity is something you practice, not just a talent you're born with. The process may feel a little uncomfortable at first, but—as the snake phobics learned—the discomfort quickly fades away and is replaced with new confidence and capabilities.

Fear of the Messy Unknown

Creative thinking in business begins with having empathy for your customers (whether they're internal or external), and you can't get that sitting behind a desk. Yes, we know it's cozy in your office. Everything is reassuringly familiar, information comes from predictable sources, contradictory data is weeded out and ignored. But out in the world, it's more chaotic. You have to deal with unexpected findings, with uncertainty, and with irrational people who say things you don't want to hear. But that is where you find insights—and creative breakthroughs. Venturing forth in pursuit of learning, even without a hypothesis, can open you up to new information and help you discover nonobvious needs. Otherwise, you risk simply reconfirming ideas you've already had or waiting for others—your customers, your boss, or even your competitors—to tell you what to do.

At the d.school, we routinely assign students to do this sort of anthropological fieldwork—to get out of their comfort zones and into the world—until, suddenly, they start doing it on their own. Consider a computer scientist, two engineers, and an MBA student, all of whom took the Design for Extreme Affordability class taught by Stanford business school professor Jim Patell. The students eventually realized that they couldn't complete their group project—to research and design a low-cost incubator for newborn babies in the developing world—while living in safe suburban California. So they gathered their courage and visited rural Nepal. Talking with families and doctors firsthand, they learned that the babies in gravest danger were those born prematurely in areas far from hospitals. Nepalese villagers didn't need a cheaper incubator at the hospital—they needed a fail-safe way to keep babies warm when they lived far from doctors who could do so effectively. Those insights led the team to design a miniature “sleeping bag” with a pouch containing a special heat-storing wax. The Embrace Infant Warmer costs 99% less than a traditional incubator and can maintain the right temperature for up to six hours with no external power source. The innovation has the potential to save millions of low-birth-weight and premature babies every year, and it came about only because the team members were willing to throw themselves into unfamiliar territory.

Another example comes from Akshay Kothari and Ankit Gupta, two students who took the d.school's Launchpad course. The class required them to start a company from scratch by the end of the 10-week academic quarter. Both were self-described “geeks”—technically brilliant, deeply analytical, and definitely shy. But they opted to work on their project—an elegant news reader for the then newly released iPad—off campus in a Palo Alto café where they'd be surrounded by potential users. Getting over the awkwardness of approaching strangers, Kothari gathered feedback by asking café patrons to experiment with his prototypes. Gupta coded hundreds of small variations to be tested each day—changing everything from interaction patterns to the size of a button. In a matter of weeks, he and Kothari rapidly iterated their way to a successful product. “We went from people saying, ‘This is crap,’” says Kothari, “to ‘Is this app preloaded on every iPad?’” The result—Pulse News—received public praise from Steve Jobs at a worldwide developer's conference only a few months later and is one of the original 50 apps in Apple's App Store Hall of Fame.

It's not just entrepreneurs and product developers who should get into the mess. Senior managers must also hear directly from anyone affected by their decisions. For instance, midway through a management off-site that IDEO held for ConAgra Foods, the executives broke away from their upscale conference rooms to explore gritty Detroit neighborhoods, where you can go miles without seeing a grocery store. They personally observed how the residents of these neighborhoods reacted to food products, and the executives spoke with an urban farmer who hopes to turn

abandoned lots into community gardens. Now, according to Al Bolles, ConAgra's executive vice president of research, quality, and innovation, such behavior is common at the company. "A few years ago," he says, "it was hard to pry my executive team away from the office. But now we venture out and get onto our customers' home turf to get insights about what they really need."

TACKLING THE MESS, RIGHT NOW

by **Caroline O'Connor** and **Sarah Stein Greenberg**

You can work up the confidence to tackle the big fears that hold most of us back by starting small. Here are a few ways to get comfortable with venturing into the messy unknown. The list gets increasingly challenging, but you can follow the first two suggestions without even leaving your desk.

- ***Lurk in online forums.*** Listen in as potential customers share information, air grievances, and ask questions—it's the virtual equivalent of hanging around a popular café. You're not looking for evaluations of features or cost; you're searching for clues about people's concerns and desires.
- ***Pick up the phone and call your own company's customer service line.*** Walk through the experience as if you were a customer, noting how your problem is handled and how you're feeling along the way.
- ***Seek out an unexpected expert.*** What does the receptionist in your building know about your firm's customer experience? If you use a car service for work travel, what insights do the drivers have about your firm? If you're in health care, talk to a medical assistant, not a doctor. If you make a physical product, ask a repair person to tell you about common failure areas.
- ***Act like a spy.*** Take a magazine and a pair of headphones to a store or an industry conference (or, if your customers are internal, a break room or lunch area). Pretend to read while you observe. Watch as if you were a kid trying to understand what is going on. How are people interacting with your offering? What can you glean from their body language?
- ***Casually interview a customer or potential customer.*** After you've gotten more comfortable venturing out, try this: Write down a few open-ended questions about your product or service. Go to a place where your customers tend to gather, find someone you'd be comfortable approaching, and say you'd like to ask a few questions. If the person refuses? No problem, just try someone else. Eventually you'll find someone who's dying to talk to you. Press for more detail with every question. Even if you think you understand, ask, "Why is that?" or "Can you tell me more about that?" Get people to dig into their own underlying assumptions.

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Fear of Being Judged

If the scribbling, singing, dancing kindergartner symbolizes unfettered creative expression, then the awkward teenager represents the opposite: someone who cares—*deeply*—about what other people think. It takes only a few years to develop that fear of judgment, but it stays with us throughout our adult lives, often constraining our careers. Most of us accept that when we are learning, say, to ski, others will see us fall down until practice pays off. But we can't risk our business-world ego in the same way. As a result, we self-edit, killing potentially creative ideas because we're afraid our bosses or peers will see us fail. We stick to safe solutions or suggestions. We hang back, allowing others to take risks. But we can't be creative if we are constantly censoring ourselves.

Half the battle is to resist judging *yourself*. If you can listen to your own intuition and embrace more of your own ideas (good and bad), you're already partway to overcoming this fear. So take baby steps, as Bandura's clients did. Instead of letting thoughts run through your head and down the drain, capture them systematically in some form of idea notebook. Keep a whiteboard and marker in the shower. Schedule daily "white space" in your calendar, where your only task is to think or take a walk and daydream. When you try to generate ideas, shoot for 100 instead of 10. Defer your own judgment, and you'll be surprised at how many ideas you have—and like—by the end of the week.

Also, try using new language when you give feedback, and encourage your collaborators to do the same. At the d.school, our feedback typically starts with "I like . . ." and moves on to "I wish . . ." instead of just passing judgment with putdowns like "That will never work." (See [chapter 18](#) of this book for more on how to give and receive feedback about creative work.) Opening with the positives and then using the first person for suggestions signals that "this is just my opinion, and I want to help"—a message that makes listeners more receptive to your ideas.

We recently worked with Air New Zealand to reinvent the customer experience for its long-distance flights. As a highly regulated industry, airlines tend toward conservatism. To overcome the cultural norm of skepticism and caution, we started with a workshop aimed at generating wild ideas. Executives brainstormed and prototyped a dozen unconventional (and some seemingly impractical) concepts, including harnesses that hold people standing up, groups of seats facing one another around a table, and even hammocks and bunk beds. With everyone coming up with these ideas, no one was scared of being judged. This willingness to consider wild notions and defer judgment eventually led the Air New Zealand team to a creative breakthrough: the Skycouch, a lie-flat seat for economy class. At first, it seemed impossible that such a seat could be made without enlarging its footprint (seats in business and first-class cabins take up much more space), but the new design does just that: A heavily padded section swings up like a footrest to transform an airline row into a futonlike platform that a couple can lie down on together. The Skycouch is now featured on a number of Air New Zealand's international flights, and the company has won several industry awards as a result.

Fear of the First Step

Even when we want to embrace our creative ideas, acting on them presents its own challenges. Creative efforts are hardest at the beginning. The writer faces the blank page; the teacher, the start of school; businesspeople, the first day of a new project. In a broader sense, we're also talking about fear of charting a new path or breaking out of your predictable workflow. To overcome this inertia, you need more than good ideas. You need to stop planning and just get started—and the best way to do that is to stop focusing on the huge overall task and find a small piece you can tackle right away.

Bestselling writer Anne Lamott expertly captures this idea in a story from her childhood. Her brother had been assigned a school report about birds, but he waited to start on it until the night before it was due. He was near tears, overwhelmed by the task ahead, until his father gave him some wise advice: “Bird by bird, buddy. Just take it bird by bird.” In a business context, you can push yourself to take the first step by asking, What is the low-cost experiment? What’s the quickest, cheapest way to make progress toward the larger goal?

Or give yourself a ridiculous deadline, as John Keefe, a d.school alum and a senior editor at radio station WNYC, did after a colleague complained that her mom had to wait at city bus stops never knowing when the next bus would come. If you worked for New York City Transit Authority and your boss asked you to solve that problem, how soon would you promise to get a system up and running? Six weeks? Ten? Keefe, who *doesn’t* work for the transit authority, said, “Give me till the end of the day.” He bought an 800 number, figured out how to access real-time bus data, and linked it to text-to-speech technology. Within 24 hours, he had set up a service that allowed bus riders to call in, input their bus stop number, and hear the location of the approaching bus. Keefe applies the same fearless attitude to his work at WNYC. “The most effective way I’ve found to practice design thinking is by showing, not telling,” he explains.

Another example of the start-simple strategy comes from an IDEO project to develop a new dashboard feature for a European luxury car. To test their ideas, designers videotaped an existing car and then used digital effects to layer on proposed features. The rapid prototyping process took less than a week. When the team showed the video to our client, he laughed. “Last time we did something like this,” he said, “we built a prototype car, which took almost a year and cost over a million dollars. Then we took a video of it. You skipped the car and went straight to the video.”

Our mantra is “Don’t get ready, get started!” The first step will seem much less daunting if you make it a tiny one and force yourself to do it *right now*. Rather than stalling and allowing your anxiety to build, just start inching toward the snake.

Fear of Losing Control

Confidence doesn’t simply mean believing that your ideas are good. It means having the humility to let go of ideas that aren’t working and to accept good ideas from other people. When you abandon the status quo and work collaboratively, you sacrifice control over your product, your team, and your business. But the creative gains can more than compensate. Again, you can start small. If you’re facing a tough challenge, try calling a meeting with people fresh to the topic. Or break the routine of a weekly meeting by letting the most junior person in the room set the agenda and lead it. Look for opportunities to cede control and to leverage different perspectives.

That’s exactly what Bonny Simi, director of airport planning at JetBlue Airways, did after an ice storm closed John F. Kennedy International Airport for a six-hour stretch—and disrupted the airline’s flight service for the next six days. Everyone knew there were operational problems to be fixed, but no one knew exactly what to do. Fresh from a d.school course, Simi suggested that JetBlue brainstorm solutions from the bottom up rather than the top down. First, she gathered a team of 120 frontline employees—pilots, flight attendants, dispatchers, ramp workers, crew schedulers, and other staff members—for just one day. Then she mapped out their disruption recovery actions (using yellow Post-it notes) and the challenges they faced (using pink ones). By the end of the day, Bonny’s grassroots task force had reached new insights—and resolve. The distributed team then spent the next few months working through more than a thousand pink Post-it notes to creatively solve each problem. By admitting that the answers lay in the collective, Simi did more than she could ever have done alone. And JetBlue now recovers from major disruptions significantly faster than it did before.

Our own experience with the open innovation platform OpenIDEO is another case in point. Its launch was scary in two ways: First, we were starting a public conversation that could quickly get out of hand; second, we were admitting that we don't have all the answers. But we were ready, like Bandura's folks with snake phobias, to take a bigger leap—to touch the snake. And we soon discovered the benefits. As of this writing, the OpenIDEO community includes about 30,000 people from 170 countries. The people involved may never meet in person, but together they've already made a difference on dozens of initiatives—from helping revitalize cities in economic decline to prototyping ultrasound services for expectant mothers in Colombia. We've learned that no matter what group you're in or where you work, there are always more ideas outside than inside. For people with backgrounds as diverse as those of Kothari, Gupta, Keefe, and Simi, fear—of the messy unknown, of judgment, of taking the first step, or of letting go—could have blocked the path to innovation. But instead, they worked to overcome their fears, rediscovered their creative confidence, and made a difference. As Hungarian essayist György Konrád once said, “Courage is only the accumulation of small steps.” So don't wait at the starting line. Let go of your fears, and begin practicing creative confidence today.

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CHAPTER 2

Train Your Brain to Be More Creative

by Bas Korsten

Great athletes train their bodies for days, weeks, and years to whip them into peak performance. Why, then, wouldn't a creator do the same with their brain? Think about it. Your brain needs fuel, and it needs to be stretched to create those "OMG!" moments on demand.

I've spent more than two decades (and counting) in the advertising industry, and contrary to popular belief, creativity isn't inherent. You have to hone it. Over time, I've figured out what I need to do to get ideas flowing freely, and much of that insight comes from my interest in neuroscience. The more we learn about the workings of our gray matter, the better we can train it, control it, and make it do what we want.

Here are a few things that have worked for me over the years.

Engaging with Nature

Many studies have shown that spending time in nature makes us more creative. Looking at trees and leaves—instead of our electronic devices—reduces our anxiety, lowers our heart rates, soothes us, and allows our brains to make connections more easily.

By spending time in nature, I'm not referring to a trek in the wilderness. Walking in an urban green space for just 25 minutes can quiet our brains and help us switch into autopilot mode. According to the *British Journal of Sports Medicine*, this state sparks our awareness of the present and fuels imagination. We are better able to connect existing notions, thoughts, and images to form a new, relevant, and usable concept.

Make disconnecting a priority. Take a walk in your neighborhood park, stroll along the beach, or just add plants to your balcony and spend some time out there. For me, walking my dog works. You'll feel the benefits of moving away from screens almost immediately.

Meditation

I know you've heard this a million times: Meditation clears our minds of jumbled thoughts, and it gives our brains the space to observe and reflect, improving task concentration and enhancing our ability to make smart decisions.

But did you know that meditation also puts the entire brain to work?

Many believe that creativity uses your right brain, and that more analytical tasks trigger the left. But in fact, neuroscientists have found that creativity draws on your *entire* brain—and meditation can give you access to it.

This intentional practice can be as simple as closing your eyes and focusing on your breath. Headspace, the popular meditation app, even has guided meditations specifically meant to inspire creativity. When we intentionally pause in awareness, we allow our minds the freedom and space to be still and creative. I practice meditation between meetings. I find a quiet space, focus on my breathing, and get my brain into an alpha state—the wakeful state of relaxation. This allows me to disconnect from my initial ideas (after all, the human brain is hardwired to take the path of least resistance) and create new pathways in my mind.

Physical Movement

Steve Jobs was a big advocate for walking meetings for a reason. Moving around has been linked to increased performance on creative tests. Exercising releases endorphins—chemicals our bodies produce to relieve stress and pain. When we are less stressed, our brains venture into more fruitful territory.

In fact, one article compared the chemical that our brain releases during physical activity to Miracle-Gro, the water-soluble plant food that helps grow bigger, healthier plants.¹ Moving around is super simple to do, especially when you're working at home. I often attend meetings while cycling on a stationary bike or plan short walks in between meetings. (You can do these in an office too.)

Try to add workout time on your calendar, and make sure not to skip it. If you feel you don't have time for a dedicated workout, block 20 minutes on your calendar and spend that time doing stretches at your desk. Find a routine that works for you.

Connections with Different Kinds of People

For deliberate sources of inspiration, not enough can be said about diversity. Remember the brain and its predisposition to take the lazy way out? Diversity makes the brain work harder by challenging stereotypes. In addition, Ernest T. Pascarella and his coauthors published an article in the *Journal of College Student Development* that showed that “exposure to diversity experiences might foster the development of more complex forms of thought, including the ability to think critically.”

I make it a point to surround myself with people who come from backgrounds different from mine, because their perspectives are a catalyst for creative thinking. Contrasting opinions spark new possibilities and allow us to make connections we hadn't seen before, leading to better decisions. There was something to be said about Abraham Lincoln's filling his cabinet with a “team of rivals.” At my agency, we've set up an “inspiration council,” which brings together our people from various regions, cultures, genders, and more, to initiate these kind of discussions.

The distributed working model has made it even easier to bring people together. I recommend using social media channels like LinkedIn and Instagram to follow and connect with people who have backgrounds and experiences that diverge from your own. Don't limit yourself by geography when you're reaching out to someone or expanding your network. We are much better at creative problem-solving when we don't have the comfort of knowing what to expect. We can become too comfortable if we only surround ourselves with people just like us. (See more about how diversity on teams improves creative performance in [chapter 15](#) of this guide.)

Use these four principles from neuroscience to give your brain the exercise that it needs. These practices will get you out of any rut. Or prevent you from getting into one in the first place.

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NOTE

1. Helen O'Callaghan, “Physical Activity Is Like Miracle-Gro: Simple Steps to Boost Your Brain as You Age,” *Irish Examiner*, March 26, 2011, www.irishexaminer.com/lifestyle/healthandwellbeing/arid-40251431.html.

Adapted from content posted on Ascend, hbr.org, June 17, 2021.

CHAPTER 3

How to Spark Creativity When You're in a Rut

by Priscilla Claman

Remember your first day at work? You were excited. There were new people to meet, new skills to be learned, new processes or products to understand.

If you are like most people, something else was different then—you. When you weren't sure or didn't understand, you asked questions, persistently. You compared what you were supposed to do on this job with what you had done in the past, and you made suggestions. You observed what your new colleagues were doing and evaluated what you saw. As a new person, you felt entitled to look at things differently and ask questions—that was a sign of your creativity.

Hiring managers look for people who can come in, assess a function, and recommend changes. They know that new people with new ideas can bring energy and creativity to a workplace, no matter what the level of their job is.

Everyone has creativity when confronted with new problems to solve or new ideas to think about. The first few months on any job can be exhausting as well as exciting, so people naturally set their work lives into a groove after a while. In time, that groove can turn into a rut. And people in a rut can develop habits that kill off their own creativity.

Are you in a creativity-destroying rut? Ask yourself these five questions:

- Is there a recurring pattern to your workdays—what you do, whom you meet with?
- Do you feel you have to agree with your colleagues and bosses to get along?
- Do you see obstacles everywhere to new ideas and new ways of doing things?
- Do you find yourself saying, “That won't work. It's been tried too many times before.”
- Do you think, “It doesn't matter what I do, really. They don't care.” Even when you're not sure who “they” are?

If you answered yes to these, you may have allowed yourself to accept thought and behavior patterns that are undermining your creativity. Of course, there may indeed be obstacles everywhere, and maybe “they” don't care, but that's not what is important. What's important is that you get those creative juices flowing again. Here are some suggestions to rekindle your creativity:

- **Think new.** Meet new people at work. Talk to new clients. Ask for new assignments. Learn something new—a new program, a new product, a new process. If you do something new every month, you won't just add to your résumé; you'll reinvigorate yourself.
- **Look for intersections.** A lot of creativity occurs at the crossroads of different people and different ideas. Look for places where your department intersects with other departments. What do they do that helps your department? What gets in the way? Volunteer for any cross-functional activity you can, whether it's a day of service or a new product team.
- **Capitalize on obstacles.** Remember that phrase “Necessity is the mother of invention”? Well, it's true. Every obstacle is an opportunity for research and analysis. Why is it there? Whom does it serve? What are its

effects? What are other ways of getting the results you're looking for? Start by selecting obstacles you can change, and move on from there. You'll build a reputation as a problem solver.

- **Share what you know.** Nothing makes you clarify your thoughts like sharing what you know, whether it's in a blog post or at a training session or as a mentor. Look for those opportunities. Volunteer. You'll be surprised at how engaged and happy these efforts make you feel.

There is an overall way of thinking about the difference between habits that sap your creativity and those that make your creativity shine. Habits that sap your creativity leave you just “doing stuff.” There's no *you* in the job, only transactions. When you develop habits that enhance your on-the-job creativity, you make the job your own. In a way, you are *above* the job, making it better, more interesting, and more effective and contributing your best self. That's the difference between someone who is just an employee and someone who is a real professional.

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Adapted from content posted on hbr.org, April 19, 2017 (product #H03M4Y).

CHAPTER 4

Schedule Your Breaks to Keep Creativity Flowing

by Jackson G. Lu, Modupe Akinola, and Malia Mason

Imagine that on a Friday afternoon, before finishing up work to start your weekend, you are asked to solve two problems that require creative thinking. Which of the following approaches do you take?

- Spend the first half of your time attempting the first problem and the second half of your time attempting the second.
- Alternate between the two problems at a regular, predetermined interval (perhaps switching every five minutes).
- Switch between the problems at your own discretion.

If you are like the hundreds of people to whom we posed this question, you would choose to switch between the two problems at your own discretion. After all, this approach offers maximum autonomy and flexibility, enabling you to change tracks from one problem to the other when you feel stuck.

But if coming up with creative answers is your goal, this approach may not be optimal. Instead, switching between the problems at a regular, predetermined interval is likely to yield the best results, according to research we published in *Organizational Behavior and Human Decision Processes*.

Why might switching at your own volition, the approach most participants in our study took, not generate the most creative outcomes? Because when attempting problems that require creativity, we often reach a dead end without realizing it. We find ourselves circling around the same ineffective ideas and don't recognize when it's time to move on. In contrast, regularly switching back and forth between two tasks at a set interval can reset your thinking, enabling you to approach each task from fresh angles.

The Research

In an experiment, we randomly assigned participants to one of the preceding three approaches. The participants who were instructed to continually switch back and forth between two problems at a fixed interval were significantly more likely to find the correct answer to both problems than were participants who switched at their own discretion or halfway through the allotted time.

A second study focused on creative ideation. In this experiment, the problems we posed had no right answers. We wanted to find out whether the benefits of stepping away from a problem at regular intervals transferred to other types of problems warranting creativity, such as brainstorming.

Again, we randomly assigned participants to one of our three task-switching approaches and asked them to generate creative ideas for two tasks. As in the first study, most people believed that they would perform best if they switched between the two idea generation tasks at their own discretion. We again found that participants who were instructed to switch back and forth between the two tasks at a fixed interval came up with the most novel set of ideas.

The issue with both other approaches seemed to be that people failed to recognize when rigid thinking crept in. Participants who didn't step away from a task at regular intervals were more likely to write "new" ideas that were very similar to the last one they had written. While they might have felt that they were on a roll, the reality was that, without the breaks afforded by continual task switching, their actual progress was limited.

The creative benefits of switching tasks have been supported by other research. For example, Steven Smith of Texas A&M University and his colleagues found that individuals instructed to list items from different categories while continually switching back and forth between the categories generated more-novel ideas than did individuals who wrote down items from one category before switching to items from the other. In a similar vein, other studies have found that brief breaks during idea generation can increase the variety of ideas that people come up with. These researchers' findings, coupled with ours, suggest that the hustle and bustle of your daily work life may facilitate your creativity if it leads you to step away from a task and refresh your thinking.

How to Schedule Your Breaks

When you're working on tasks that would benefit from creative thinking, consciously insert breaks to refresh your approach. Set them at regular intervals—use a timer if you have to. When it goes off, switch tasks: Organize your reimbursement receipts, check your email, or clean your desk, and then return to the original task. If you're hesitant to break away, because you feel that you're on a roll, be mindful that this feeling might be a false impression. We tend to generate redundant ideas when we don't take regular breaks; ask yourself whether your latest ideas are qualitatively different. Finally, don't skip your lunch breaks, and don't feel guilty about taking breaks, especially when you are feeling stuck. Doing so may actually be the best use of your time.

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Adapted from "To Be More Creative, Schedule Your Breaks" on hbr.org, May 10, 2017 (product #H03MY1).

CHAPTER 5

Get Your Brain Unstuck

by Ron Friedman

Your colleagues logged off hours ago, but you're still staring at the screen, waiting for a breakthrough. You haven't produced anything substantive for hours. Yet for reasons you can't understand, it's been impossible to walk away. Even now, the answer seems *so close*.

If your work involves creative thinking, you are bound to encounter times like this—times when you feel stuck. Perhaps you're not sure how to start a project, respond to a client email, or structure an upcoming presentation. You're trying to be productive, yet as you turn the problem over in your head again and again, you find yourself running into the same barriers.

When this happens, a common reaction is to redouble your efforts. Who doesn't love a good persistence story? Most of us have been taught that the only thing standing between us and a successful outcome is hard work.

But the research tells us something different. While grit does have its role, when it comes to creative solutions, dogged persistence can actually backfire.

A funny thing happens when you're thinking about a problem. The more time you spend deliberating, the more your focus narrows.

It's an experience familiar to all of us. When you first encounter a problem, certain solutions immediately burst to the fore. Occasionally, none of them seem quite right, so you try to reexamine the issue, giving it a fresh look and then another. Before you know it, something counterproductive happens. You lose sight of the big picture and become fixated on details. And the harder you try, the less likely it is that unexpected, novel ideas will enter your train of thought. At this point, you've reached a point of diminishing returns on your efforts.

So what should you do?

Research suggests that when you find yourself at an impasse, it's often fruitful to use *psychological distance* as a tool. By temporarily directing your attention away from a problem, you allow your focus to dissipate, releasing its mental grip. When you let go, loose connections will suddenly appear, making creative insights more likely.

While most of us intuitively know that a three-day weekend or an extended vacation can yield a renewed perspective, those options aren't always available, especially when we're facing a deadline.

But that doesn't mean that we can't reap the benefits of psychological distance in our day-to-day work. Here are three practices that can help.

Struggling for more than 15 minutes? Switch tasks

Task-switching can be perilous. When we're making good progress, allowing distractions to hijack our attention can derail our focus. But the moment we experience ourselves getting stuck, the rules shift. Here, a well-timed distraction can be a boon to creativity.

When we let go of a problem, our perspective expands. This phenomenon explains why we discover so many solutions in odd places, like the shower, the commute back home, or the visit to the gym. Redirecting our attention to an unrelated task also provides room for incubation, a term psychologists use to describe nonconscious thinking. Studies show that after a brief distraction, people generate a greater number of creative solutions to a problem than do people who spend the same period focusing on it intently.

The trick is to recognize when you are feeling stuck and resist the temptation to power through. In many cases, it is precisely when we are at our most discouraged that we can derive the greatest benefit from walking away.

For tasks requiring creative thinking, schedule multiple sessions over several days

Often, the most productive way of resolving a difficult problem is to alternate between thinking about it very deeply and then strategically shifting your attention elsewhere. Instead of setting aside one continuous block of time to work on a creative project, schedule shorter, more frequent sessions. By planning multiple periods of deep thinking, you're guaranteed a few transitions away from your task, ensuring that your focus expands.

Put mind-wandering periods to good use

Creative solutions rarely emerge when we're in the office. Which is why it can be helpful to keep an ongoing list of "thinking problems" that you can access on the go. Glance at your list just before entering mind-wandering periods, like when you're going out for a sandwich or traveling between meetings. A new context can lead to a fresh perspective.

Ultimately, the key to harnessing the power of psychological distance involves accepting that often, the best ideas don't appear when we're pushing ourselves to work harder. They prefer sneaking up on us, the moment we look away.

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CHAPTER 6

Four Ways to Spark Creativity When You're Feeling Stressed

by Susan Peppercorn and Maren Gube

The human brain comes equipped with powerful response mechanisms to protect us during uncertain times. In disruptive or unpredictable situations, the brain tries to interpret incoming stimuli and form a plan of action if the news is bad. When precarious circumstances are prolonged and combined with anxiety and fear, the brain's biological response is to hide rather than explore. From an evolutionary point of view, survival ends up taking precedence over creativity, as the stress hormones that ready us for battle also interrupt the networks involved in creativity.

The cognitive load from this subconscious scanning and scenario-building is one reason that we may feel tired and uninspired—and less creative—when going through a stressful time (or living in a stressful time). Why is an awareness of these effects important? By understanding how the brain functions when we're under threat or dealing with potentially frightening challenges, we find it easier to design strategies to alleviate both the conscious and unconscious fears that arise in the face of disruption, thereby supporting our own creativity. Here are four strategies that incorporate this understanding of your brain's functionality to nourish and feed your creative instincts.

Reasserting Control

Experiencing a loss of control is one of the main symptoms of uncertainty and ambiguity. Continued uncertainty can provoke feelings of being adrift and make it hard to plan or make decisions. Under these conditions, your brain's emotional amygdala circuits become increasingly active: Instead of choosing proactive and thoughtful actions through the prefrontal cortex, you may become reactive and more likely to allow fear to guide you. Fear is the enemy of creative thinking because it directs neural resources toward neutralizing the perceived threat.

That's why it can be a good idea to reduce uncertainty by planning your own disruptions. "When we're the disruptors, it's still scary," says Jonathan Fields, host of the *Good Life Project* podcast. "But we have a sense of agency and a powerful connection to a reason. We don't know how it will end or if we'll be able to do what we set out to do, but at least we've chosen the direction, and we're heading toward our ultimate destination." According to Fields, this approach can center you enough so that you can maintain a sense of control, breathe more easily, and think more clearly. You feel these benefits because focusing on your own ultimate destination engages your prefrontal cortex, which, as the seat of planning and control, helps regulate your thoughts and emotions—and ultimately the behaviors that can lead you to either open or close your creative spigots.

Designing a Conducive Environment

Having clarity, direction, and the right mental tools reduces stress and fear of the unknown while giving the brain freedom to make creative connections rather than dealing with imminent threats. Yet remote and hybrid work have complicated that picture, making it more important to be intentional about how work, communication, and collaboration opportunities are structured. Without the chance office conversations that sometimes spark creativity, companies that care about product innovation need a way for their people to stay consistently creative, even at a distance.

Andrew Yanofsky, head of operations at toy manufacturer WowWee, explains how hybrid and remote work has forced teams to be more thoughtful and structured about the yields expected from brainstorming meetings. His company has made changes to reflect that reality. “Since we can’t rely on face-to-face conversations for ideation,” he says, “we have become more intentional about setting mandates for meetings, moderating the sessions, and monitoring the steps post-brainstorm. Someone is designated as the ‘pen holder’ for the whole process and keeps an eye on key deliverables from beginning to end.” Although many people view creativity as a free form of outside-the-box thinking, Yanofsky explains that by “making a sandbox and then playing in it,” his teams have formalized that process while ensuring that they provide psychological safety for contributors.

Exercising New Mental Muscles

As described, you’re keeping the prefrontal cortex engaged and your fear at bay by taking ownership of your own disruption and clarifying how you and your teams approach creative processes. You can further step up creativity during stressful periods by cross-training your brain and immersing yourself in new and different activities. Research has shown that venturing outside your primary area of expertise can enhance creative thinking.

For example, one study revealed that diversifying your areas of focus can boost cognitive flexibility, which allows for more fluid ideation and creativity. Social media marketing consultant and composer Marie Incontrera chose this route and credits collaborating with Dorie Clark, an HBR contributor and one of Thinkers50’s top business thinkers, as vital to sparking her creativity in a different direction from her core work. During the pandemic, Incontrera and Clark created a new work of musical theater, a lesbian spy musical called *Absolute Zero*, which was incubated in a prestigious development program run by Apples and Oranges Arts. Their goal is to bring the show to Broadway in five years.

Increasing Connection

Both individuals and organizations struggle with the creativity-killing effects of prolonged stress, disruption, and uncertainty. One solution to help our brain overcome these challenges is to facilitate the release of the “social hormone,” oxytocin, in our brains. Lourdes Olvera-Marshall, a diversity and inclusion strategist and an executive coach, told us that her team works toward this by allowing time at the beginning of a Zoom meeting for non-task-related conversation so that colleagues can connect in a personal way. She also continues to make time to connect one-on-one with her team members at least every other week with no agenda. “I’ve seen how these small moments generate connection and psychological safety as team members feel more comfortable with each other,” Olvera-Marshall explains. “Plus, the meetings are more fun.”

This type of social connection becomes a powerful creativity booster in part because it increases mutual trust, a foundational element for creativity. Olvera-Marshall has seen a significant increase in idea sharing when a few meetings a week start with positive, personal interactions. “Everyone approaches work more relaxed, which activates the parasympathetic system and generates the right climate for new, diverse ideas, which in turn boosts motivation and creativity,” she says.

By employing these strategies, Olvera-Marshall is helping her company’s leaders foster not just individual creativity but also team creativity through socio-emotional support. This approach improves the brain’s neurochemical balance by reducing stress hormones and increasing pro-social hormones, helping to reconnect the creativity networks that continuous uncertainty may have disrupted.

With each of the proactive steps just described, you can turn the stress of disruption into fuel for creativity, lessening your evolutionary fear reflexes and instead nourishing and revitalizing your creative brain circuits. Implement these ideas any time your innovative juices need a boost but particularly during your toughest challenges. After all, periods of rapid change, disruption, and global uncertainty are arguably the times when bold, brave creativity is needed the most.

Susan Peppercorn is an executive career transition coach and speaker. She is the author of *Ditch Your Inner Critic at Work: Evidence-Based Strategies to Thrive in Your Career*. Numerous publications, including the *New York Times*, the *Wall Street Journal*, *Fast Company*, the *Boston Globe*, and *Self* magazine, have tapped her for career advice. **Maren Gube** is Executive Director of Resiliti. She guides organizations pursuing cultures of fearless collaborative creativity through her research and professional practice. Her award-winning work on creativity and why women leave STEM fields has earned citations on both sides of the Atlantic.

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CHAPTER 7

You Know How to Focus—Learn How to Unfocus

by Srini Pillay

The ability to focus is an important driver of excellence. Focused techniques such as to-do lists, timetables, and calendar reminders help people to stay on task. Few would argue with that, and even if they did, there is evidence to support the idea that resisting distraction and staying present have benefits: Practicing mindfulness for 10 minutes a day, for example, can enhance leadership effectiveness by helping you become more able to regulate your emotions and make sense of past experiences. Yet as helpful as focus can be, there's also a downside to focus as it is commonly viewed.

The problem is that excessive focus exhausts the focus circuits in your brain. It can drain your energy and make you lose self-control. This energy drain can also make you more impulsive and less helpful. As a result, decisions are poorly thought-out, and you become less collaborative.

So what do we do then? Focus or unfocus?

In keeping with recent research, both focus and unfocus are vital. The brain operates optimally when it toggles between focus and unfocus, allowing you to develop resilience, enhance creativity, and make better decisions.

When you unfocus, you engage a brain circuit called the *default mode network*. Abbreviated as DMN, we used to think of it as the “do mostly nothing” circuit because it only came on when you stopped focusing effortfully. Yet, even when the mind is “at rest,” this circuit uses 20% of the body's energy.

The DMN needs this energy because it is doing anything but resting. Under the brain's conscious radar, this circuit activates old memories; goes back and forth between the past, present, and future; and recombines different ideas. Using this new and previously inaccessible data, you develop enhanced self-awareness and a sense of personal relevance. And you can imagine creative solutions or predict the future, thereby leading to better decision-making too. The network also helps you tune into other people's thinking, thereby improving team understanding and cohesion.

RUN AN ENERGIZER TO SPARK A NEW IDEA

by Duncan Wardle

What are you doing when you get your best ideas? Showering? Jogging? Commuting? I've worked with thousands of individuals around the world, and I've never heard anyone say “working.”

How do you get into that relaxed brain state on demand? By running an *energizer*: a quick exercise that can get you and your team members (or classmates or friends) to laugh and relax. This practice frees your brain up from tiresome executive functions like weekly meetings and reports and gives you access to your own unrelated knowledge and experience that can uncover new ideas.

Energizers come in all shapes and sizes. “Storyteller” is one example. Participants pretend they’re the expert of some unique field (think a leading designer of parachutes for elephants) and are interviewed by other subjects in the room on their “expertise.” Another example is “Heroes and Villains.” In this exercise, a group is split into two teams, and each is asked to give a pitch for (heroes) or against (villains) a particular idea, such as hosting a Pet Olympics.

Of course, listening to a fictitious expert on elephant parachutes or adults debating a Pet Olympics invariably leads to laughter, and I’ve found that playfulness (in the form of laughter) triggers relaxation. In this state, the parts of your brain that allow a stray thought, a random memory, or an image to surface can combine in novel ways.

Although it would be unfeasible to be this playful all the time, energizers are a great way to bring in structured moments of play during team meetings, in the classroom, or just in your daily life as a way to unlock the parts of your brain that sit idle throughout the day.

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Adapted from “So, You Think You’re Not Creative,” Ascend, on hbr.org, March 29, 2021.

There are many simple and effective ways to activate this circuit in the course of a day.

Use positive constructive daydreaming

Positive constructive daydreaming (PCD) is a type of mind-wandering different from slipping into a daydream or guiltily rehashing worries. When you build this practice into your day deliberately, it can boost your creativity, strengthen your leadership ability, and reenergize the brain. To start PCD, you choose a low-key activity such as knitting, gardening, or casual reading, then wander into the recesses of your mind. But unlike slipping into a daydream or guilty, dysphoric daydreaming, you might first imagine something playful and wishful—like running through the woods or lying on a yacht. Then you swivel your attention from the external world to the internal, with this image in mind while still doing the low-key activity.

Studied for decades by psychology professor Jerome Singer, PCD activates the default mode network and metaphorically changes the silverware that your brain uses to find information. While focused attention is like a fork—picking up obvious conscious thoughts that you have—PCD commissions a different set of silverware. It employs a spoon for scooping up the delicious mélange of flavors of your identity (the scent of your grandmother, the feeling of satisfaction with the first bite of apple pie on a crisp fall day) and chopsticks for connecting ideas across your brain (to enhance innovation). PCD also uses a marrow spoon for getting into the nooks and crannies of your brain to pick up long-lost memories that are a vital part of your identity. In this state, your sense of self is enhanced—which, according to business scholar Warren Bennis, is the essence of leadership. This sense of self is your psychological center of gravity, a grounding mechanism that helps you boost your agility and manage change more effectively too.

Take a nap

In addition to building in time for PCD, leaders can also consider authorized napping. Not all naps are the same. When your brain is in a slump, your clarity and creativity are compromised. Studies show that after a 10-minute nap, you become much clearer and more alert. But if you have a creative task in front of you, you will probably need a full 90 minutes for more complete brain refreshing. Your brain requires this longer time to make more associations and dredge up ideas that lie in the recesses of your memory network.

Pretend to be someone else

When you're stuck in a creative process, unfocus may also come to the rescue when you embody and live out an entirely different personality. Educational psychologists Denis Dumas and Kevin Dunbar found that people who try to solve creative problems are more successful if they behave like an eccentric poet rather than a rigid librarian. Given a test in which they have to come up with as many uses as possible for any object (for example, a brick), those who behave like eccentric poets have superior creative performance. This finding holds even if the same person takes on a different identity.

When in a creative deadlock, try this exercise of embodying a different identity. This exercise is likely to get you out of your own head, and it allows you to think from another person's perspective. I call this practice *psychological halloweenism*.

For years, focus has been the venerated ability among all abilities. Since we spend 46.9% of our days with our minds wandering away from a task at hand, we crave the ability to keep it fixed and on task. Yet if we built PCD, 10- and 90-minute naps, and psychological halloweenism into our days, we would probably preserve focus for when we need it, and we'd use it much more efficiently too. More importantly, the ability to unfocus will allow us to update information in our brains, giving us access to deeper parts of ourselves and enhancing our agility, creativity, and decision-making.

Srini Pillay, MD, is the cofounder of Reulay and the CEO of NeuroBusiness Group. He is also a technology innovator and entrepreneur in the health and leadership development sectors and an award-winning author. His latest book is *Tinker, Dabble, Doodle, Try: Unlock the Power of the Unfocused Mind*. He was formerly a part-time assistant professor at Harvard Medical School and has taught in the Executive Education Programs at Harvard Business School and Duke Corporate Education.

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CHAPTER 8

Look Outside Your Industry for More-Creative Solutions

by Bill Taylor

A big challenge in times of disruption and uncertainty is for people and organizations to keep learning as fast as the world is changing—to analyze problems they haven’t encountered before and to make sense of opportunities they haven’t thought about before.

That’s why leaders should encourage their colleagues to learn from experts in fields they’ve never worked in. Practices that are routine in one industry can be revolutionary when they migrate to another, especially when they challenge conventional wisdom in that industry. What better way to fuel your company’s imagination than to look for inspiration outside your field? If you want to learn fast, learn from strangers.

In their book *Benchmarking for Best Practices*, Christopher E. Bogan and Michael J. English share a business case study that illustrates how accepted ideas from one field can quickly transform another field. “In 1912,” they write, “a curious Henry Ford watched men cut meat during a tour of a Chicago slaughterhouse. The carcasses were hanging on hooks mounted on a monorail. After each man performed his job, he would push the carcass to the next station. When the tour was over, the tour guide said, ‘Well, sir, what do you think?’ Mr. Ford turned to the man and said, ‘Thanks, son, I think you may have given me a real good idea.’ Less than six months later, the world’s first assembly line started producing magnetos in the Ford Highland Park Plant.”

Or consider a more timely case study: A *Wall Street Journal* article by Ben Kesling described how the chaplain of an overwhelmed hospital in Chicago helped nurses deal with the mental and emotional fallout of the early Covid-19 wave and the rise of the Omicron variant. This chaplain, who was also an army veteran, “noticed that phrases nurses were using in conversation sounded like what he had heard from troops who had served in combat zones.” So, to help nurses stay strong as they waged war on the virus, he borrowed concepts and techniques developed by the army to help troops deal with the trauma of war. “He could actually draw that parallel between us and people that have been veterans of war,” one nurse marveled about the chaplain and his program. “I never even made that connection because for me—I’ve been a nurse for 20 years on this unit—I’ve never seen this kind of trauma to our team ever. So when he made that connection, I was like, oh my gosh, how have I never realized that?”

Take another example: Several years ago, London’s Great Ormond Street Hospital, a children’s hospital renowned for its cardiac care, was struggling with poorly designed handoffs when it transferred patients from one step of a complex medical procedure to the next. So Martin Elliott, head of cardiac surgery, and Allan Goldman, head of pediatric intensive care, studied high-powered professionals who, despite being from a totally unrelated field, were better than anyone else was at organizing handoffs—the pit crew of Ferrari’s Formula One racing team.

The doctors and the pit crew worked together at the team’s racing center in Italy, at the British Grand Prix, and in the hospital’s operating room. Members of the pit crew were struck by how clumsy the hospital’s handoff process was, not to mention the fact that it often lacked a clear leader. (In Formula One races, a so-called lollipop man wields an easy-to-see paddle and calls the shots.) Moreover, they noted how noisy the process was. Ferrari pit crews operate largely in silence, despite (or because of) the roar of engines around them. Thanks to the techniques the medical staff learned from these outsiders—techniques that were accepted wisdom in racing circles—the hospital redesigned its handoff procedures and sharply reduced medical errors.

Learning from strangers doesn't always have to be about dealing with an unfamiliar situation or solving a specific problem. Colonel Dean Esserman, during his transformational tenure as chief of police in Providence, Rhode Island, pushed his insular department to open itself up to original ideas, and new ways of thinking. One of his initiatives was a fabulously creative program called Cops and Docs. Esserman's detectives regularly sat in with doctors at Brown University Medical School as the physicians discussed tough cases. The detectives watched and listened as the doctors analyzed clues about a patient (symptoms), sorted through evidence (test results), and identified the culprit (disease).

In turn, doctors sat in on the police department's command meeting to learn how cops dealt with conflicting and confusing information, ruled out suspects, and cracked their cases. Esserman's goal was for his department to "become a place that embraces research, that figures out and spreads methodologies that work in ways that medical schools do." For set-in-their-ways detectives to learn new perspectives on policing, their chief understood they had to learn from experts in a field unrelated to policing.

These times of rapid change and instability have created all kinds of questions for leaders and organizations. One of the biggest questions is this: Do we have new ideas about where to look for new ideas? When it comes to innovation and problem-solving, there will always be a place for old-fashioned, time-consuming R&D—research and development.

Today, though, there is also a place for a different kind of R&D—rip off and duplicate. The fastest way for organizations to make sense of challenges they are seeing for the first time is to survey unrelated fields for ideas that have been working for a long time. Why gamble on untested strategies and insights if you can quickly apply strategies and insights that are already proven elsewhere? That's how leaders can help their colleagues keep learning as fast as the world is changing.

Bill Taylor is the cofounder of *Fast Company* and the author, most recently, of *Simply Brilliant: How Great Organizations Do Ordinary Things in Extraordinary Ways*. Learn more at williamctaylor.com.

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